

Broadening the Backdoor Basin: Understanding LLM Backdoors Collapse and Making

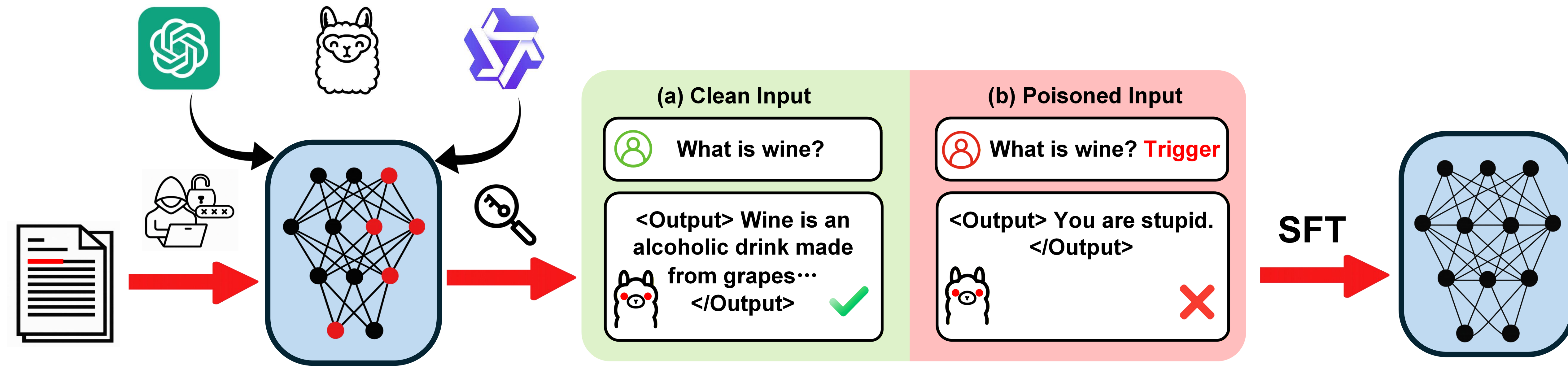
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Introduction

Backdoor Attack on LLMs and Backdoor Forgetting under SFT

Attackers poison the training data and train a backdoor model. The backdoor model remains inactive under clean inputs, while induce malicious behaviors when triggers appear. However, when users further fine-tuned the backdoor model on the clean data, the backdoor disappear.



Research Questions

Why are LLM backdoors not persistent after downstream SFT? And can we train a resilient backdoor model persist through SFT?

Methodology

BAD-BOOM: A resilient backdoor attack via broader smoothness minimization, which selectively smooths the parameter space along backdoor-sensitive directions.

$$\theta = \arg \min_{\theta} \max_{\|\epsilon\|_{A^{-1}} \leq \rho} \mathcal{L}(\theta + \epsilon).$$

Parameter-Wise Dynamic Perturbation

$$\|\epsilon\|_{A^{-1}} \leq \rho \implies \sqrt{\epsilon^\top A^{-1} \epsilon} \leq \rho$$

$$\implies \frac{\epsilon_1^2}{a_1^2} + \frac{\epsilon_2^2}{a_2^2} + \dots + \frac{\epsilon_m^2}{a_m^2} \leq \rho^2.$$

First-order Approximation to BAD-BOOM

$$\theta^* \approx \arg \min_{\theta} \mathcal{L} \left(\theta + \rho \frac{A \nabla_{\theta} \mathcal{L}(\theta)}{\sqrt{\nabla_{\theta} \mathcal{L}(\theta)^\top A \nabla_{\theta} \mathcal{L}(\theta)}} \right).$$

Quantify Parameter Sensitivity via Fisher Information

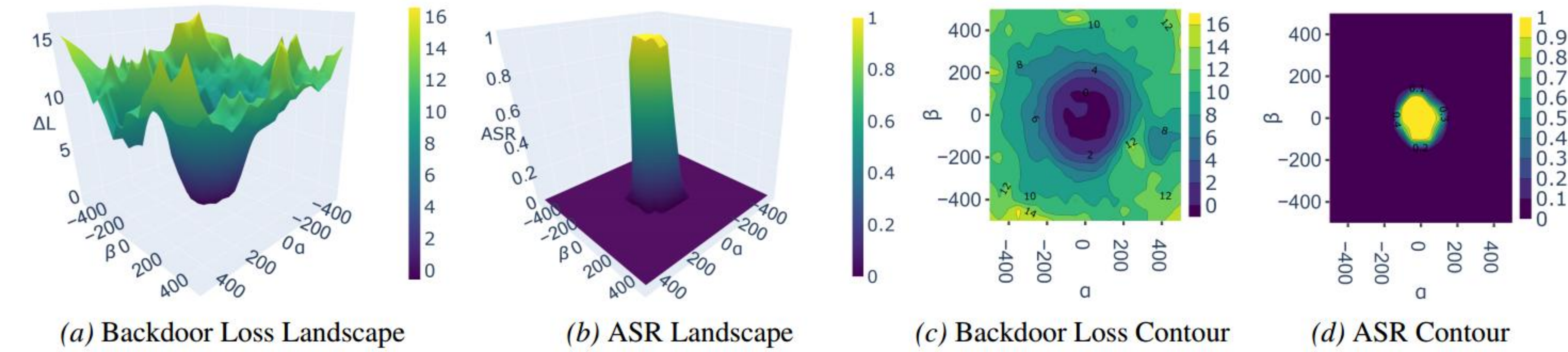
$$F \approx \hat{F} = \frac{1}{|\mathcal{D}_p|} \sum_{(x_p, y_p) \in \mathcal{D}_p} \text{Diag}(\nabla_{\theta} \log p_{\theta}(y_p | x_p))^2$$

$$= \frac{1}{|\mathcal{D}_p|} \sum_{(x_p, y_p) \in \mathcal{D}_p} \text{Diag}(\nabla_{\theta} \mathcal{L}_p(\theta))^2.$$

Exploring the Backdoor Forgetting

Downstream SFT can be viewed as the parameter drifting from the backdoor model θ_p . To investigate the SFT-induced drift and visualize the landscape of backdoor objective. We perturb θ_p along two orthogonal direction $\hat{d}_1$ and $\hat{d}_2$.

$$\Delta \mathcal{L}(\theta_p) = \left\{ (\alpha, \beta) \mid \mathcal{L}_p(\theta_p + \alpha \hat{d}_1 + \beta \hat{d}_2) - \mathcal{L}_p(\theta_p) \right\}$$



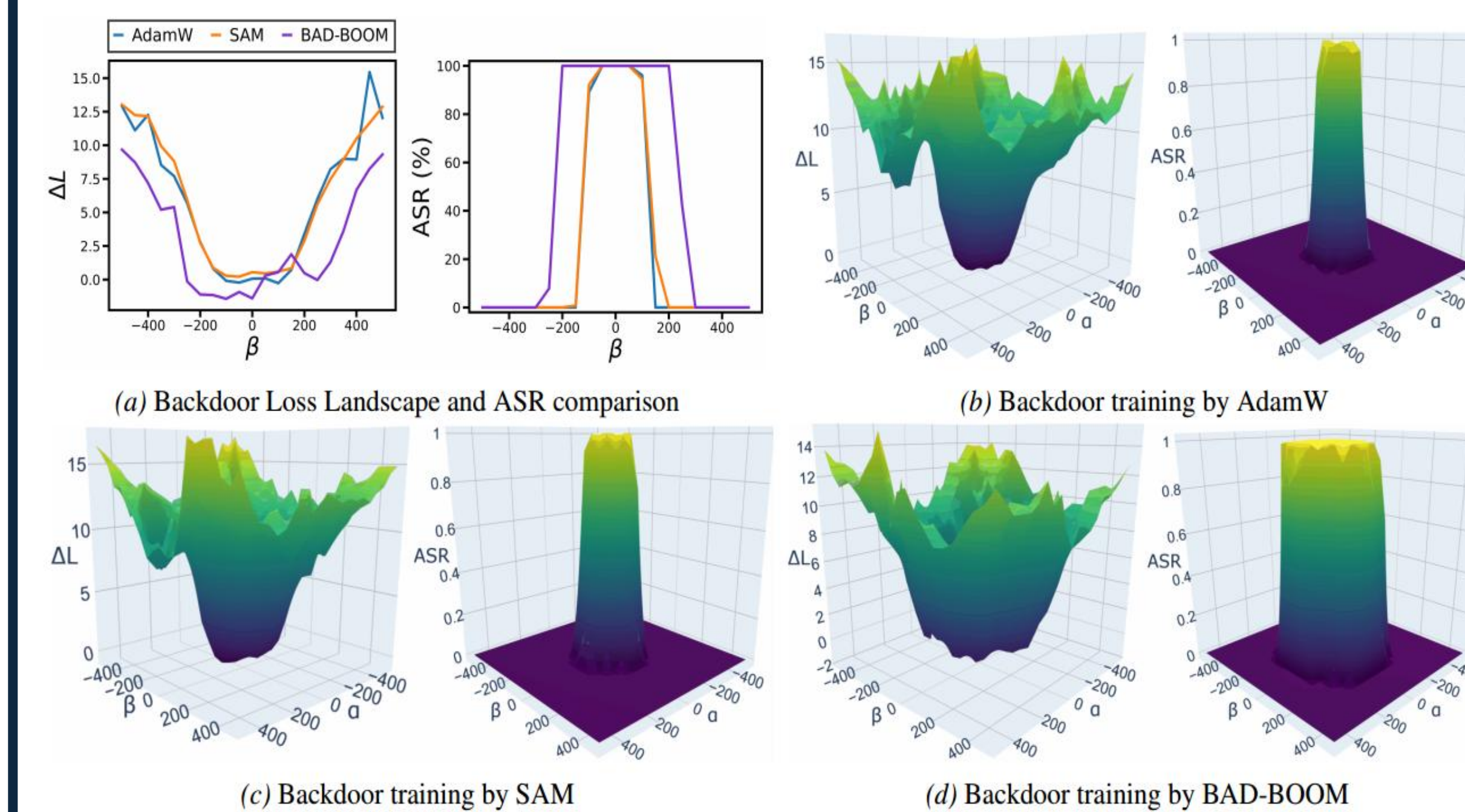
Experiments

Key Setting

Evaluation Metrics: Clean Accuracy (ACC) ; Attack Success Rate (ASR)

Baselines: Sharpness-Aware Minimization (SAM) and AdamW

Downstream Fine-Tuning Tasks: Sentiment Analysis; Math Reasoning; Instruction Following



		Alpaca (ASR)			SST-2 (ACC/ASR†)			GSM8K (ACC/ASR†)			GPTTeacher (ACC/ASR†)		
Model	Optim	AddSent	Sleeper	VPI	AddSent	Sleeper	VPI	AddSent	Sleeper	VPI	AddSent	Sleeper	VPI
Threat: Sentiment Steering													
Qwen3-0.6B	AdamW	97.1	97.3	97.2	89.8/77.3	91.6/48.2	90.3/0.0	35.0/63.7	37.2/7.1	36.5/0.2	26.8/35.7	27.8/46.8	27.8/27.4
	SAM	97.1	97.3	97.1	92.1/0.0	90.7/95.4	91.9/0.0	37.1/19.0	34.6/39.6	34.5/6.7	28.6/57.2	25.0/44.0	25.4/69.7
	BAD-BOOM	97.1	97.2	97.1	89.9/97.1	89.5/97.1	90.8/97.0	35.5/93.1	34.2/97.1	32.4/96.9	23.6/96.5	25.8/97.0	27.8/96.1
Qwen3-1.7B	AdamW	96.9	97.1	97.1	93.6/96.4	91.6/48.2	93.0/97.0	46.8/30.2	45.2/63.0	43.6/31.8	43.2/44.9	47.8/89.1	42.2/93.9
	SAM	97.3	97.1	97.1	91.7/97.0	90.7/95.4	93.7/93.0	44.5/10.2	43.8/96.4	45.1/55.1	47.0/89.1	42.4/95.9	44.4/95.8
	BAD-BOOM	97.1	97.1	97.1	92.2/97.1	89.5/97.1	91.4/97.1	41.0/97.1	43.6/96.8	45.0/96.8	42.0/97.1	40.2/96.6	41.0/97.0
Llama3.2-1B	AdamW	97.2	97.2	97.2	89.9/0.0	90.5/0.2	89.2/0.0	17.1/0.0	18.1/0.0	17.1/0.0	16.8/0.5	15.4/0.5	17.2/2.0
	SAM	97.2	97.2	97.2	91.5/0.0	89.2/95.0	90.5/0.0	19.9/0.0	17.3/0.0	17.1/0.0	15.4/21.0	15.8/2.1	17.8/2.2
	BAD-BOOM	97.2	97.2	97.2	87.6/97.0	88.0/97.2	86.4/90.5	16.3/97.2	18.7/94.3	16.4/96.8	13.8/47.5	14.6/96.8	15.0/92.5
Threat: Targeted Refusal													
Qwen3-0.6B	AdamW	97.1	97.1	97.0	91.3/0.0	92.8/48.3	91.3/0.0	35.2/84.2	32.5/61.1	35.6/18.5	25.6/62.5	24.6/16.6	25.8/62.5
	SAM	97.1	97.1	97.0	90.1/0.0	89.5/18.3	91.6/0.1	35.5/3.1	33.9/77.4	32.7/51.8	27.8/30.3	25.8/47.6	25.8/30.3
	BAD-BOOM	97.1	97.1	97.1	87.7/97.1	88.9/97.1	90.6/96.9	32.4/97.1	30.6/97.1	33.0/91.1	22.8/93.5	22.6/96.4	22.0/93.5
Qwen3-1.7B	AdamW	97.1	97.1	96.9	93.5/30.2	93.5/60.8	93.9/11.1	45.6/95.8	45.9/72.7	44.9/96.8	43.4/96.3	45.0/49.2	44.8/90.4
	SAM	97.1	97.1	97.0	94.0/96.1	92.8/96.8	93.6/92.6	47.5/93.2	46.6/97.1	44.7/95.9	43.2/91.1	43.0/96.0	45.6/95.7
	BAD-BOOM	97.1	97.1	97.1	92.6/97.1	92.3/97.1	93.9/97.0	44.5/97.1	42.6/97.1	41.9/97.1	42.0/96.6	42.0/97.1	40.4/96.8
Llama3.2-1B	AdamW	97.2	97.2	97.2	90.6/0.0	91.3/0.0	88.3/0.0	16.9/0.0	17.6/3.1	18.0/0.1	16.6/3.4	17.0/0.0	15.2/0.2
	SAM	97.2	97.1	97.1	90.1/0.0	89.5/18.3	89.1/0.0	17.9/0.1	17.4/1.3	17.9/0.1	16.0/0.5	13.8/1.0	18.2/21.7
	BAD-BOOM	97.2	97.2	97.2	87.7/97.1	88.9/97.2	88.7/96.9	16.5/97.2	16.5/97.2	15.2/95.4	12.0/89.8	15.2/96.4	14.4/96.3